A Novel Hybrid, BERT and Deep Learning Model Network Intrusion Detection System for Healthcare Electronics

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***Abstract*—IoMT has become an attractive playground for cybercriminals due to its market value and rapid growth. As the amount of sensitive data transmitted across IT infrastructures grows, healthcare organizations and companies that generate wearable data become targets for attackers. Recently, government agencies and healthcare organizations have prioritized collecting this data and using machine learning to protect users' privacy. In all test cases, the proposed method performed better than any other method, including the IoMT intrusion dataset, with an accuracy increase of 2.9%. It can also monitor IoMT networks within the healthcare and medical environment to protect IoMT devices and networks from attackers. Based on the ECU-IoHT dataset, it achieves 99% performance improvement in accuracy, precision, recall, and F1-score compared with existing anomaly detection models. The proposed model shows higher detection accuracy than the existing latest state-of-arts.**

***Index Terms*—IoMT, wearable electronics devices, detection systems.**

I. Introduction

MOBILE devices and wearables have accelerated Internet of Things (IoT) technologies in healthcare. In addition to real-time human activity monitoring, wearables and mobile sensors can enhance medical rehabilitation and elderly care. Therefore, gain insight into people's daily behaviours and interactions using Human Activity Recognition (HAR) in ubiquitous computing environments. These developments have led to extensive research on the Internet of Healthcare Things (IoHT) [1], [2]. In real-time, wireless sensor networks can collect and transmit posture data (e.g., accelerometers and gyroscopes) from various body parts (e.g., heads, chests, upper arms, forearms, shins, etc). (For instance, wearable devices such as inertial sensors and smartphones are examples.) [3]. As a result of sensor technologies, it may be possible for HAR to become more robust when it comes to sensing and fusion of multimodal data, enabling us to create applications and services based on real-time big data environments that enhance sensed information for human-centred applications and services [4], [5], [6].

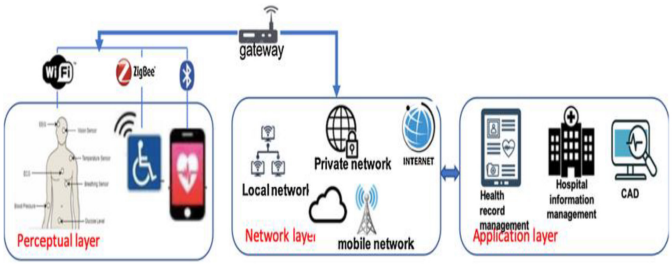

Fig. 1. The architecture of the IOMT system.

It adheres to a three-tier architectural framework for IoT applications that comprises perception, network, and transmission layers as a focused personification of IoT in the medical area. As shown in Figure 1, the IoMT architecture is based on the cloud.

Wearable's and mobile devices have built-in inertial sensors, which can easily gather motion data from people's physical activities. It is common to use smart devices to recognize, analyze, and understand the status of a person across a variety of different environments in a wide range of applications and systems [7], [8], [9]. HAR relies heavily on sensor data, but several challenges can be overcome. It is essential to design well-labelled datasets to extract features that are effective in machine learning methods [10]. Although people may not use smart devices to perform sensible actions, built-in sensors on those devices continuously generate ADL data. As a result, annotating and recording well-labelled data is labour-intensive. Data collected by wearable sensors that are weakly labelled

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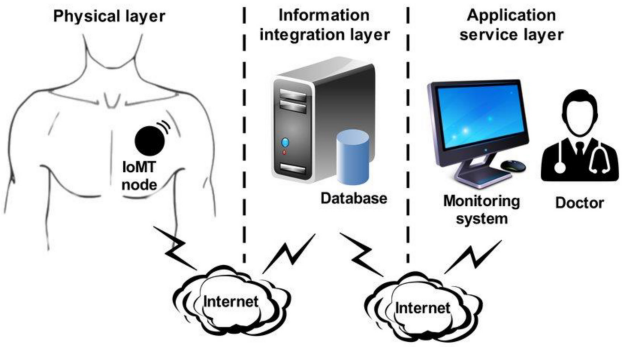

Fig. 2. The IOMT network.

refers to data that is mostly unlabelled, with only a few segments labelled [11], [12], [13]. Hence, a weakly supervised or semisupervised learning framework is the best way to handle such situations. The diagram in Figure 2 illustrates an example of medical devices connected to a hospital network.

Today's Machine learning is often called privacy-preserving [14], [15]. To circumvent privacy concerns, Google [16] proposes federated learning [17], which leverages collaborative learning across IoT devices (Internet of Things). Due to various constraints, IOT applications are difficult to implement using traditional federated learning and central servers. As a result, all IoT devices will have a significant decline in accuracy as a result of a crash or malicious modification of the global model by the central server (e.g., Edge server) [18], [19], [20], [21], [22], [23], [24]. The federation of learning is hampered by the lack of power in IoT devices. Due to this resource limitation, federated learning must be implemented with optimized energy consumption [25].

For example, federated learning has been highly successful in optimizing mobile edge networks, suggesting and predicting Google keyboard queries [26], [27], detecting COVID-19 [28], [29], [30], facilitating vehicle communication [31], [32], the Internet of Drones [33], Intrusion detection, Augmented Reality, [34], [35], [36] etc. Often, researchers have difficulty deciding which type of learning method will be most advantageous for testing and evaluating their proposed security methods in IoT applications (e.g., centralized or federated learning). It makes selecting the right method of federated deep learning crucial. Taking advantage of the new capabilities in federated deep learning, we plan to conduct an experimental study with a comprehensive approach to better understand IoT security through federated deep learning.

Information and networks are protected from cyber-attacks through cyber security. An IoHT attack detection method must be developed [37]. Due to safety risks and the potential for harm and death, these attack datasets are not publicly available [38]. Our methodology addresses these risks using a novel ECU-IoHT dataset [38]. Various cyberattacks are reflected in this report. Based on these issues, this paper presents a deep learning-based system to detect an extensive range of cyberattacks. Unlike existing approaches, this system can detect ARP spoofing, Denial of Service (DoS) attacks, Nmap port scans, and Smurf attacks in the Internet of Things [38]. Considering the types of attacks observed in the IoHT environment, this deep learning approach appears to be the most effective method of multiclass classification in the IoHT environment to date.

This paper contributes the following contributions

- It is more important to detect different types of attacks than specific attacks in the IoHT environment using a hybrid model using BERT and deep learning.
- The proposed method trains and evaluates the model using the ECU-IoHT dataset in the healthcare domain. Since many publicly available datasets are inappropriate for health care, the ECU-IoHT dataset was selected.
- The accuracy of the intrusion detection rate achieved by BERT and Deep Learning-based intrusion detection systems was 99.11% for five frequently occurring intrusions.

The paper is organized into five sections. Work related to the topic is presented in Section II. Section III presents how the model is mathematically analyzed and calculated. A detailed description of the experimentation procedures, results analysis, and performance evaluation can be found in Section IV. The paper is concluded in Section V.

II. Literature Review

Recent theories suggest that using new technologies for mutual benefit is crucial to economic growth. This definition explains why Japanese Society 5.0 refers to an organization that uses technologies and organizations to improve the world through connected objects, big data, and artificial intelligence [39], [40]. We live in a changing world in this new era of globalization and technological advancement. The evolution also directly affects vertical markets and society. Values also change with the diversity and complexity of citizens and the environment. Digital technologies are at the heart of Industry 4.0, Made in China 2025, and Society 5.0 revolutions are happening worldwide. The wave of digital transformation is driving all of these activities.

The industrial strategy is flattering and reliant on digital transformation [41], [42]. Due to this transformation, most industrial applications and companies have completely transformed. New work models require employees who understand and can use them. Hence, mastering process digitization, redefining business lines, and integrating data within the organization are the pillars of this transformation.

Several recent proposals have proposed how patients can be monitored and assessed in real-time [43], [44], [45], [46]. A framework for detecting and evaluating stress is presented [46]. Based on patient vitals, breathing rate, heart rate, and average blood pressure, stress signals are first detected using skin conductance parameters and then estimated stress levels using fuzzy inference systems. A study [43] proposes a framework for managing healthcare data and making informed decisions. An algorithm is in place to detect emergencies and send critical records directly to the coordinator. Sensors are sampled at different frequencies according to ANOVA and Fisher tests, which adjust the sampling rate based on patient health conditions. A fuzzy set theory-based decision matrix is proposed at the coordinator for data fusion

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and decision-making. This framework has many disadvantages despite its advantages when it comes to patient assessment and monitoring: 1) A medical staff cannot prevent patients from entering dangerous situations since sensors only transmit critical records to the coordinator; 2) the hospital will not archive any data for low critical patients, so doctors cannot review patient archives to monitor patient progress; 3) sensor rate cannot be adapted for patients with medium criticality based on ANOVA or Fisher tests, as they are sensitive only to significant data variation.

A. Risks of the Internet of Medical Things (IOMT)

Healthcare is one of the industries where IoT is being implemented, which comes with many risks:

1) If a patient's personal information is disclosed, it can seriously impact the patient's health and hospital reputation.
2) When any medical device manipulates the transmitted data, it will cause the wrong medical description and high drug dosage due to the falsification of the data [47].
3) It has been reported that employees who are unhappy in their jobs, those who are connected to organized crime, or those who have bribed leak confidential information of patients, which puts the patient's life and privacy at risk.
4) In some cases, doctors and nurses may cause permanent disabilities to their patients, or worse, even cause their death due to a lack of training.
5) Some specialized robots can perform medical operations that are inaccurate, which might result in the patient's life being seriously affected or even causing them to die [48].

The Internet of Medical Things offers the safety, reliability, dynamic capabilities, and scalability of traditional Internet of Things and larger capabilities. A combination of the Internet of Things and the "Internet of Medical Things" can solve medical and chronic problems for the elderly and chronic diseases, which are sufficiently generic to cover a wide range of diseases that require hydrogenous monitoring and actuation simultaneously. The Internet of Medical Things faces a major challenge in standardization. Vendors must take security measures to prevent hacking. Standardization of medical devices is essential to ensuring their compatibility [49]. IOMT is not just about mobility but also about monitoring and providing medical care to patients wherever they are, regardless of their mobility. Healthcare systems worldwide are becoming increasingly aware that they need to be updated with the latest technological innovations, and developing countries are providing them with solutions to improve healthcare.

IoT has many opportunities and challenges in the medical field, especially in using and developing connected and distributed medical devices [50]. With IOMT, body sensors can be easily networked, allowing retrieval of patient data from distant locations without installing several biosensors and consuming many resources. IoMT focuses more on wearable devices than personal medical devices because wearable devices often take wearable forms. In addition to sensors and wearables, medical devices, clinical devices, and other medical devices comprise the Internet of Medical Things ecosystem. Since the medical community adheres to strict ethical standards, there are some concerns that biomedical devices need to address. In this regard, there are

1) *Reliability:* The reliable system should achieve its functional goals every time, so there should be no unexpected failures under normal circumstances. For IoT systems to provide guaranteed information that is collected, The potential diagnostic nature of the test requires reliability.
2) *Safety:* Safe schemes should not harm the operating system. As far as IoT in medical devices is concerned, at the very least, it must demonstrate that the system will not harm its users.
3) *Security:* All medical systems must be protected from external threats and attacks because they collect highly sensitive information about individuals [51].

B. State-of-the-Art of Existing Work

An asymmetric (public key) algorithm differs from a symmetric (secret key) algorithm. The security provided by asymmetric cryptosystems is better than that of symmetric encryption, but it requires significantly greater computing power. Symmetric encryption is less resource-intensive than asymmetric cryptosystems, which provide a higher level of security. Most data exchanged between healthcare and personal servers occurs over public channels like the Internet, so stronger security measures should be implemented.

Public-key cryptography is frequently used in the cloud for authentication, data storage, and access control (Table I). In comparison with traditional algorithms like Rivest-Shamir-Adleman (RSA), ECC offers the most security [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62], [63], [64]. Considering how lightweight they are for devices with limited resources, symmetric cryptographic algorithms have been used in access control and data transmission between and between IoMT sensors. A symmetric cryptographic algorithm is often used as a session key in hybrid security schemes [55], [61], [65], [66]. Moreover, the most commonly used attacks in adversarial/security analysis are chosen plaintext attacks, replays, impersonations, insider attacks, and MitM attacks. Their security schemes have been analyzed in many research studies, including Mutual Authentication (MA), Forward Security (FS), Contextual Privacy (CP), Anonymity and Traceability (A&T), and Unlinkability. Separately, those who used computer simulations, the rest performed experiments on a real-time basis [57], [65]. While some used computer simulations, others used actual hardware.

III. Methodology

If an adversary modifies a URL, it will have a standard structure. In literature, URLs are analyzed as regular sentences by exploiting their standard structure. Hence, NLP techniques can be used to analyze URLs (e.g., to predict words and classify words). Several NLP techniques rely on word embeddings for their effectiveness. There are several methods for embedding words, but BERT is more successful than the others. Google AI researchers have developed an open-sour BERT, an open-source model developed by Google AI researchers ce,

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TABLE I
IoMT HEALTHCARE SYSTEM SECURITY AND PRIVACY RESEARCH (EHR = ELECTRONIC HEALTH RECORD)

Method & Authors	Data Type	Simulation	Prototype	Adversarial-analysis	Purpose	Methodology
Public-key (A. Zhang et al., 2016)	PHI/EHR	Yes	No	Privacy and Contextual Information (CP), Mutual Authentication (MA), Anonymity and Traceability (A&T), Ability to Unlink	Data transmission	Certificates less generalized sign-encryption
Public-key (Y. Yang et al., 2017)	PHI/HER	Yes	No	Chosen keyword-CPA	Access control, data retrieval	Pairing-based cryptography
Symmetric (H. Huang et al., 2017)	PHI/HER	No	Yes	Eavesdropping, CPA, Replay, Camouflage Trust Attack	Data transmission	Group send-receive model
Public-key (Mehmood et al., 2018)	-	Yes	No	Unlinkability, A&T, Denial of Service (DoS), Traffic Analysis, Revocation, MA, Man-in-the-Middle (MitM), Replay	Cloud-based authentication	Elliptic curve cryptography
Public-key (Vaniprabha & Poongodi, 2019)	PHI/HER	Yes	No	Sinkhole, wormhole, Sybil	Access control,	Elliptic curve cryptography
Public-key (X. Liu & Ma, 2018)	-	Yes	NO	Key compromise impersonation, A&T, know-key attack, Replay, MA, Impersonation, Internal, MitM	Cloud-based authentication	Elliptic curve cryptography
Public-key (Chaudhary et al., 2018)	PHI/HER	No	No	MA, MitM, quantum computing, known ciphertext attack, CPA, data masquerading, replay.	Cloud-based authentication data encryption, Access control,	Lattice-based cryptography
Public-key (Luo et al., 2018)	PHI/HER	No	No	Data leakage and destruction, collusion, insider attacks	Data storage, transmission, access control	Slepian-Wolf-coding-based secret sharing
Hybrid (Elhoseny et al., 2018)	Medical images	Yes	No	-	Data transmission	RSA, Rivest, Shamir, and Adleman (AES) are the most widely used cryptographic standards
Symmetric (Y. Zhang et al., 2018)	PHI/HER	Yes	No	Collusion, attribute privacy	Access control,	Encryption based on an attribute-based ciphertext policy
Symmetric (Gupta et al., 2019)	-	Yes	No	Impersonation. Eavesdropping, FS, insider, password guessing, A&T, node capture, MitM	Wearable device authentication	Bitwise XOR, a one-way hash function
Hybrid (Ostad-Sharif et al., 2019)	-	Yes	No	Password guessing, node capture, replay, impersonation, DoS, session key verification	Cloud-based authentication	Bitwise XOR, one-way hash function Elliptic curve cryptography
Public-key (Ding et al., 2019)	PHI/HER	Yes	No	Identity privacy, authenticity, unforgeability	Cloud-based data storage, Access control,	Diffie-Hellman key exchange, identity-based cryptosystem
Public-key (Boussada et al., 2019)	PHI/HER	Yes	No	Unlink ability, know-key attack, CPA, CP, A&T, replay, forging, unobservability, time correlation, mitigating, pseudonymity	Data transmission	Elliptic curve cryptography
Hybrid (Y. Yang et al., 2019)	-	Yes	No	CPA	Cloud-based Access control,	Encryption based on attributes instead of passwords
Hybrid (Cheng et al., 2019)	-	Yes	Yes	Whether the two-way identification can be achieved through MA, replay, and MitM remains to be seen.	Device/node authentication	Elliptic curve cryptography

and a pre-trained NLP model called BERT [68]. Data representation efficiently uses autoencoders and word embeddings through projections onto appropriate vector spaces. Word embedding models, similar to autoencoding, learn vector space embeddings. Figure 3 shows the BERT architecture, including procedures for pre-training and fine-tuning.

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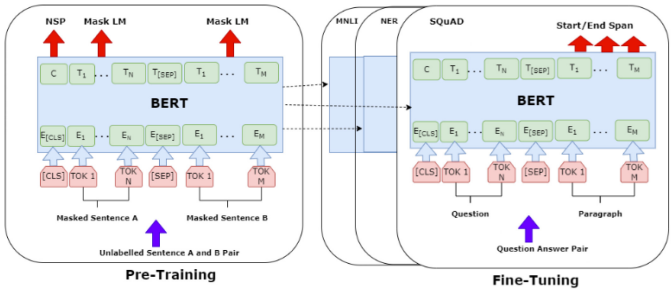

Fig. 3. Model for BERT's pre-training and fine-tuning.

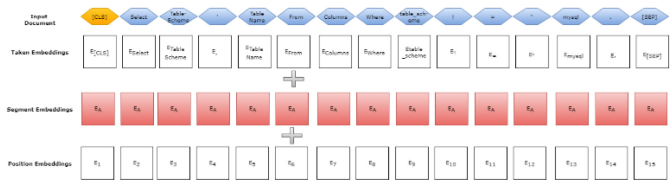

Fig. 4. Tokenization of URLs.

An encoder-decoder stack can be viewed as BERT. On the other hand, traditional encoder-decoder architectures forget some learned characteristics, especially those accumulated over time. Data processing left-to-right and right-to-left is more efficient with transformers than with traditional encoder-decoder architectures. The BERT transformer network consists of encoders and decoders, including feed-forward and self-attention mechanisms.

It is solved by inserting tokens at the beginning and end of sentences (CLS and SEP tokens). Text processing also uses a maximum length setting. If the sentence is shorter than the maximum length, zeros are entered for the empty fields. Alternatively, the length will be cut off if it exceeds the maximum. An index is eventually created after dividing a sentence into tokens. We will only consider the hidden attention associated with the first token to simplify things. After that, we determine the vector distance in the word space for each word. Figure 4 illustrates our framework, which begins with tokenizing the text via the BERT tokenizer. Several schemes exist for classifying texts.

These models can analyze context in both directions through extensive pre-training on huge corpora. Deep constituent networks are difficult to generalize when the inner layers do not retain this ability. A hyper-parameter-based model is then created based on the fine-tuned outer layers of the network. The hyper parameter and their values are presented in the Table II.

A WordPieceModel is used to tokenize and manage words that are not in the standard vocabulary [69]; the dictionary uses sub-words common to most words. As an alternative, the special token [SEP] separates phrases, and the hidden dimension H is represented by the special token [CLS].

In the classification layer, transformers represent the final fully connected layer, represented by the last hidden layer; we calculate the probability for each of the K categories by using $C \in R^H$ as a vector and $W \in R^{KxH}$ as a matrix.

$$P = sofmax\left(CW^T\right) \tag{1}$$

TABLE II
HYPER-PARAMETERS

Hyper Parameter	Value
Attention heads	12
Batch size	8
Epochs	5
Hidden size	768
Hidden layers	12
Learning Rate	0.00001
Maximum Sequence Length	512
Parameters	110M

The BERT model provides a Categorical Cross Entropy loss function by default, but an alternative is provided in this article. Torch library's Binary Cross Entropy (BCE) loss function was chosen as the loss function in this case; predicting single labels with this method is the best. The BCE with Logits method was selected to achieve greater numerical stability. This method combines a BCE with a sigmoid using the function LogSumExp. It is possible to use BCEwL for batch size N in the following way:

$$\begin{aligned} l(x, y) &= L = \{l_1, \ldots, l_N\}^T, \ l_n \\ &= -w_n\{y_n.\sigma(x_n) \\ &\quad +(1 - y_n).\log(1 - \sigma(x_n))\big] \end{aligned} \tag{2}$$

Transformer: BERT architecture relies heavily on transformers [70]. Additionally, [CLS] is placed before [X], while [SEP] follows after [X] and [Y], starting from the sequence of sub-words [X] and [Y]. Thus, embedding occurs through the embedding function E and the normalization layer LN:

$$\widehat{h}_i^0 = E(x_i) + E(i) + E(1_x) \tag{3}$$

$$\widehat{h}_{j+|x|}^0 = E(y_i) + E(j + |x|) + E\left(1_y\right) \tag{4}$$

$$\widehat{h}^0 = Dropout\left(LN\left(\widehat{h}^0\right)\right) \tag{5}$$

The element-wise Gaussian error linear units (GELU) activation function and the multihead self-attention (MHSA) function originate from the Feed Forward layer transform blocks.:

$$\widehat{h}^{i+1} = skip\left(FF, Skip\left(MHSA, \ h^i\right)\right) \tag{6}$$

$$Skip(f, h) = LN(h + Dropout(f(h))) \tag{7}$$

$$FF(h) = GELU\left(hW_1^T + b_1\right)W_2^T + b_2 \tag{8}$$

The new position and $h^i \in R^{(|x|+|y|)\times d_h}$, $W_1 \in R^{4d_h \times d_h}$, $b_1 \in R^{4d_h}$, $W_2 \in R^{4d_h \times d_h}$, $b_2 \in R^{4d_h}$, are:

$$\begin{aligned} \left[\ldots, \widehat{h}_i, \ldots\right] &= MHSA\left(h_1, \ldots, h_{|x|+|y|}\right) \\ &= W_0 Concat\left(h_i^1, \ldots, h_i^N\right) + b_0 \end{aligned} \tag{9}$$

Attention heads N are characterized by the following:

$$h_i^j = \sum_{k=1}^{|x|+|y|} Dropout(\propto_k^{(i,j)} \ W_v^j h_k \tag{10}$$

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TABLE III
THE ECU-IOHT DATASET INCLUDES THE FOLLOWING ATTACKS AND INPUT FIELDS

Attack types	Instances	Input fields used
ARP Spoofing	2359	5
DoS	639	5
Nmap Port scan	6836	5
Smurf Attack	77920	5
No Attack	23454	5
Total instance	111207	

$$\propto_k^{(i,j)} = \frac{exp\frac{\left(W_Q^j h_i\right)^T W_k^j h_k}{\sqrt{d_h/N}}}{\sum_{k'=1}^{|x|+|y|} exp\frac{\left(W_Q^j h_i\right)^T W_k^j h_{k'}}{\sqrt{d_h/N}}} \tag{11}$$

where $h_i^j \in R^{(\frac{d_h}{N})}$, $W_0 \in R^{d_h \times d_h}$, $b_0 \in R^{d_h}$ *and* $W_Q^j, W_K^j, W_V^j \in R^{d_h/N \times d_h}$,

To conduct Web attacks, attackers usually modify URLs, so it is essential to maintain the string of words and characters. Our framework then converts tokenized URLs into numerical values merged into word vectors, represented as words or sentences. The architecture of our system is consequently built on a text classification infrastructure. Using the BERT tokenizer, as in many text classification schemes, we begin by tokenizing the text, as shown in Figure 4.

IV. RESULTS ANALYSIS AND DISCUSSION

Detection of DDoS attacks in medical IoT networks was tested using our collaborative learning model against conventional models. We compared our simulation results to those of other models to validate our proposed model's generalization in different environments. We implemented our proposed model with an NVIDIA GeForce RTX 3060 GPU on a Windows server with an Intel Core i7-10700F CPU and 32GB of RAM.

A. Dataset and Feature Presentation

ECU-IoHT, which analyses cyberattacks on the Internet of Things [38], was used for the experiment. There are eight fields in the ECU-IoHT dataset: the host address, the source address, the destination address, the network protocol, the length of packets, and information about packets.

An attack label in the dataset can be divided into type (network status) and type (attack type). Network status fields were used to determine labels. Packet information reports do not help detect attacks, so we excluded them from the input features. To accomplish this, the training process used five input features. In Table III, shows the ECU-IOHT dataset statistics.

Port scanning and DoS attacks are both part of DDoS attacks. As another form of DDoS attack, Smurf attacks flood targeted network devices with Internet Control Message Protocol (ICMP) packets to turn off network services and render them unusable. In this experiment, we used Nmap port scanning, DoS attacks, and Smurf attacks. Thus, a total of 108849 attacks were recorded during this experiment, consisting of 85395 attacks and 23454 no-attacks.

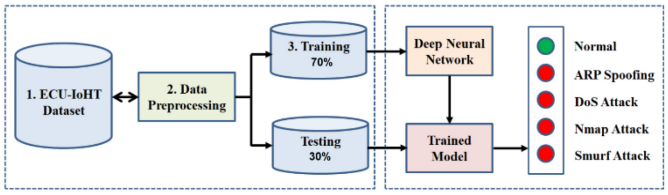

Fig. 5. A description of the dataset's layout and features.

TABLE IV
AN INSTANCE OF AN ATTACK IS DIVIDED INTO A TRAINING INSTANCE AND A TEST INSTANCE

Attacks	Attack instances	Training instances (70%)	Training instances (30%)
DoS	639	447	192
Nmap Port scan	6836	4785	2051
Smurf Attack	77920	54544	23376
No Attack	23454	16418	7036
Total instance	108849	76194	32655

B. Dataset Pre-Processing

Categorical information is encoded in the dataset as the first step in pre-processing. Target output was encoded using label encoding, while source IP, destination IP, and network protocol were encoded using One-Hot encoding. We then used min-max scaling to normalize the dataset.

$$X_{new} = \frac{X - X_{min}}{X_{max} - X_{min}} \tag{12}$$

The dataset contains the field X. There are two groups of input fields. Time stamps and packet lengths are in one group, while the remaining fields are in another. A group of input fields was assigned to each collaborative agent. ARP Spoofing, Nmap attacks, Smurf attacks and DoS attacks are among the cyber-attacks presented in Figure 5.

As part of the implementation process, the following stages were followed: (i) various cyber-attacks are analyzed using the ECU IoHT dataset; (ii) this dataset contains five categorical features encoded using one-hot encoding; (iii) to prepare for the multiclass classification process, the dataset is categorized as Normal, DoS attacks, Nmap attacks, ARP Spoofing and Smurf attacks; (iv) there is a 70% training dataset and a 30% testing dataset; (v) to train the proposed model, these labels are selected as target features, and a trained model is provided from the training dataset; (vi) The training dataset is used to test the proposed model for its ability to predict normal attacks or other types of attacks. Table IV summarizes the training and testing instances of 108849 attack instances from the ECU-IoHT dataset.

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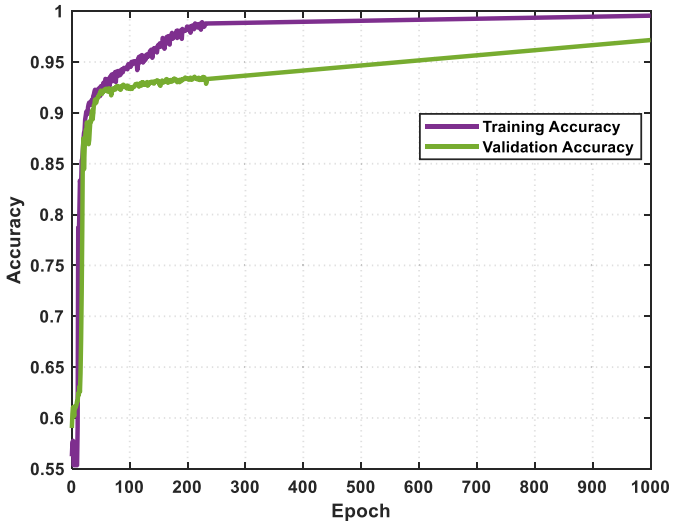

Fig. 6. Training and validation accuracy versus number of epochs.

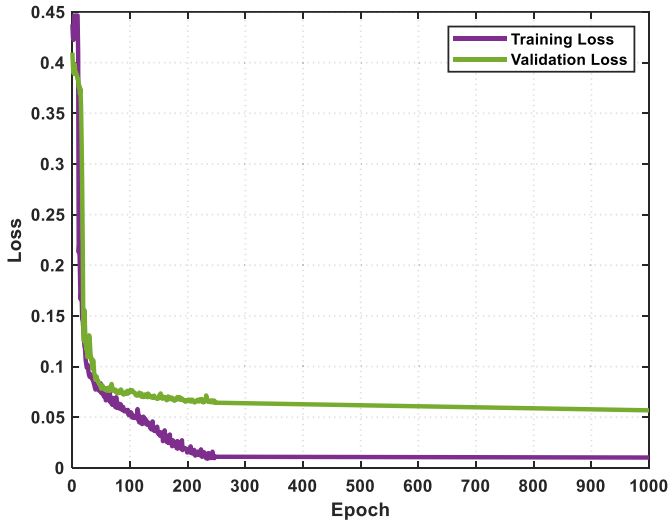

Fig. 7. Training and validation loss versus number of epochs.

C. Evaluation Metrics

Accuracy, Precision, Recall, and F1-Score are evaluated to detect anomalies. We evaluate the proposed anomaly detection model using Accuracy, Precision, Recall, and F1-Score:

Accuracy: A measure of how well the model detects log sequences.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{13}$$

Precision: Percentage of anomalies identified by the model that are correctly detected.

$$Precision = \frac{TP}{TP + FP} \tag{14}$$

Recall: The percentage of anomalies that the model correctly detects out of all the anomalies.

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{15}$$

True positives (TP) represent anomalies detected correctly by the model. The model detects TN (true negative) when there are normal log sequences. An anomalous log sequence may be mistakenly detected as a true anomaly by the model if there are several false positives (FP). An anomaly not detected by the model is a false negative (FN).

D. Experimental Results

Based on a dataset containing normal and abnormal instances, an approach is proposed to detect different attacks in IoHT. Different epoch values between 500 and 1000 are used to train the model. The proposed model's 500 and 1000 epochs training time is 1582 seconds and 2559 seconds, respectively. In this model, 1000 epochs were used for maximum validation accuracy, and the minimum loss of 1000 epoch dataset is shown in Figures 6 and 7.

Following the proposed model, we compare the F1 score for ARP Spoofing, Nmap, Smurf attacks, Port Scan, and DoS attacks. ARP and Nmap attacks of all types can be detected, as described in Figure 8.

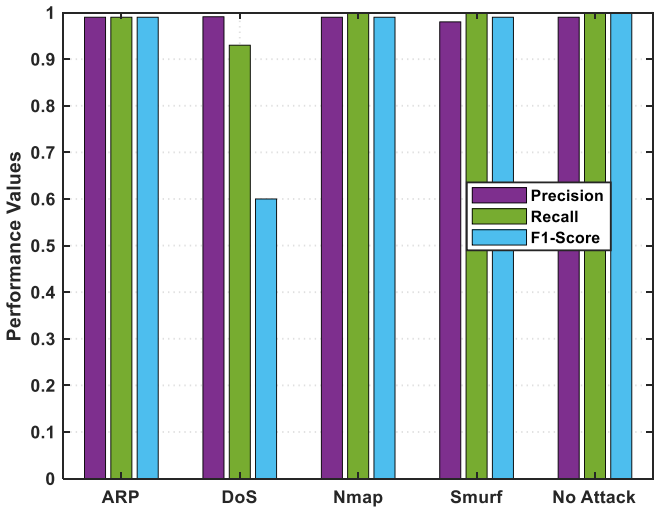

Fig. 8. Performance Analysis based on the feature classes with 1000 epochs.

As shown in Figure 9, the proposed model performed better with 1000 epochs than it did with 500 epochs in terms of Accuracy, Precision, Recall, and F1 Score. Regarding Accuracy, Average Precision, Recall, and F1-Score, the model with 1000 epochs performs better than the others.

In this sense, the proposed technique is superior to other techniques and can detect normal class attacks and other attacks, such as ARP spoofing and Nmap attacks, as illustrated in Figure 8.

A comparison of the proposed system against existing models is shown in Table V. Based on Table V, it can be seen that the system proposed is significantly more effective at detecting Nmap PortScan attacks than the existing system.

V. Conclusion

A security compromise in the healthcare system could directly result in the loss of human life, which makes the IoMT (Internet of Medical Things) require increased security.

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TABLE V
THE COMPARATIVE PERFORMANCE EVALUATION WITH THE EXISTING MODEL

		ARP	DoS	Nmap	Normal	Smurf
Proposed Model	Precision	0.99	0.99	0.99	0.98	0.99
	Recall	0.99	0.93	1	1	1
	F1 Score	0.99	0.60	0.99	0.99	1
Author (Manimurugan et al., 2020)	Precision			0.96	0.96	
	Recall			0.96	0.98	
	F1 Score			0.97	0.97	
Author DM Model (Diro & Chilamkurti, 2018)	Precision		0.97	0.98	0.99	
	Recall		0.99	0.99	0.97	
	F1 Score		0.98	0.98	0.98	
Author SM Model (Diro & Chilamkurti, 2018)	Precision		0.96	0.87	0.99	
	Recall		0.99	0.99	0.95	
	F1 Score			0.93		
Author (Anthi et al., 2018)	Precision			0.97		
	Recall			0.97		
	F1 Score			0.97		

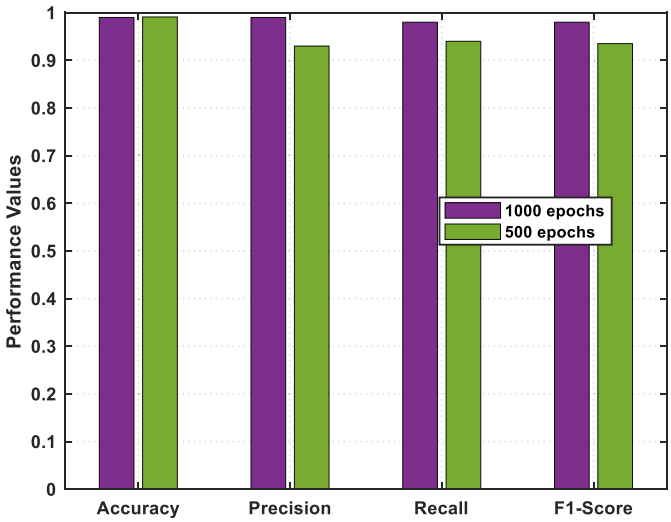

Fig. 9. Performance Analysis based on the accuracy, precision, recall and F1-Score with 500 and 1000 epochs.

This article proposes a hybrid model for detecting intrusions in IoMT systems based on network flows and patient biometrics that combines BERT and deep learning. It passes network flows and biometric information into multiple hidden deep-learning layers to learn the optimal feature representation. As part of BERT, centralized healthcare operations will be transformed into distributed healthcare operations while maintaining user privacy. The proposed model's precision, recall and F1-Score are 99%, indicating that the positively identified intrusions are correct. A similar performance is observed for the recall and F1-score, which vary from 98-99% based on the ARP, DoS, Nmap, Normal, and Smurf, respectively. A comparison with the state-of-the-art study showed that the model was more reliable and robust. Future work could also improve and test the method presented in this paper. Various datasets, including many classes, features, and many others, will also be tested with the proposed methodology. A larger dataset must be used to validate this model, and low-complexity feature sets can enhance its effectiveness. The performance of this model can also be enhanced by using transformer models and incremental learning operations to optimize the model continuously in real-time clinical settings.

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